

Ticket Upgrade, Exchange & Conversion

Functional Design & Backend Screens

TICVAI | Professional Reference Document

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Board 1 — Final Screen Register

Area 11 — Ticket Upgrade, Exchange & Conversion

Board 1 — Upgrade & Conversion Management

10 Backend Configuration Screens — Detailed SOW

I recommend keeping Area 11 as **1 board / 10 screens**. Unlike Dynamic Pricing, the complexity here can be contained cleanly in one board as long as we separate **eligibility, conversion paths, financial treatment, operational execution, and audit** properly.

The key architecture should be:

Existing Ticket / Entitlement → Eligibility Check → Allowed Upgrade/Conversion Path → Financial Recalculation → Validation → Payment/Refund if applicable → New Entitlement/Credential → Original Ticket Invalidation → Audit

And we should avoid duplicating other TICVAI engines:

- **Area 10 Pricing Engine** calculates price differences, tax, fees and rounding.
- **Ticketing Catalogue** owns ticket/product definitions.
- **Access Control** owns credential validity and access synchronization.
- **CRM** provides customer/member context.
- **Payment Engine** processes additional payment/refund.
- **Area 11** owns the business rules determining **whether, when and how one ticket can become another.**

11.1.1 — Upgrade & Conversion Command Center

Purpose

Provide administrators and operations teams with one centralized view of all ticket upgrade, exchange and conversion configurations and operational activity.

KPI Cards

Display:

- Active Upgrade Paths

- Active Conversion Rules
- Products Eligible for Upgrade
- Products Excluded
- Upgrades Today
- Conversions Today
- Upgrade Revenue
- Pending Exceptions
- Failed Conversions
- Expiring Rules
- Configuration Conflicts
- Manual Overrides

Configuration Directory

Each configuration displays:

- Rule ID
- Rule Name
- Source Product
- Target Product
- Transaction Type
- Venue
- Channel
- Effective Period
- Financial Method
- Approval Requirement
- Status
- Owner

Transaction Types

Support:

Upgrade

Standard → Premium

Downgrade

Premium → Standard

Exchange

Event A → Event B

Conversion

Day Ticket → Annual Pass

Person-Type Conversion

Child → Adult

Product Conversion

General Admission → Combination Ticket

Operational Activity

Example:

426 upgrades completed today

AED 84,600 incremental revenue

17 conversions rejected by eligibility rules

4 manual exceptions awaiting approval

Filters

Filter by:

- Venue
- Event
- Product
- Transaction Type
- Channel
- Customer Segment
- Rule Status
- Effective Date

Acceptance Condition

Authorized users can monitor the complete upgrade and conversion configuration portfolio and operational activity from one workspace.

11.1.2 — Upgrade & Conversion Path Builder

Purpose

Define exactly which products/tickets may be converted into which other products.

This becomes the central **conversion relationship engine**.

Path Structure

Configure:

Source Product

→

Target Product

Examples

Standard Admission

→ Premium Admission

1-Day Pass

→ 2-Day Pass

Single Attraction

→ Multi-Attraction Pass

Day Ticket

→ Annual Membership

Child Ticket

→ Adult Ticket

Standard Seat

→ VIP Seat

Path Configuration**Capture:**

- Path Name
- Path Code
- Source Product
- Target Product
- Transaction Type
- Venue
- Event
- Performance
- Effective From
- Effective To
- Allowed Channels
- Customer Segment
- Status

Direction**Configure:****One-Way**

Standard → Premium

or:

Bidirectional

Product A ↔ Product B

where commercially permitted.

Multiple Targets

One source product may have several allowed paths.

Example:

Standard Ticket →

- Premium

- VIP
- Fast Track
- Annual Pass

Chained Upgrade

Configure whether:

Standard → Premium → VIP

is allowed.

Circular Path Detection

TICVAI should detect inappropriate configurations such as:

A → B → C → A.

Visual Relationship Map

Administrators should be able to visualize product conversion paths as a relationship diagram.

Acceptance Condition

Administrators can explicitly define and visualize all permitted upgrade, downgrade, exchange and conversion relationships between TICVAI products.

11.1.3 — Upgrade Eligibility & Qualification Rules

Purpose

Determine whether a particular ticket/customer/transaction qualifies for a configured upgrade or conversion path.

A path existing does **not** automatically mean every ticket can use it.

Eligibility Conditions

Rules can evaluate:

- Ticket Status
- Product
- Ticket Type
- Event
- Performance
- Visit Date
- Timeslot
- Customer Type
- Customer Segment
- Membership
- Loyalty Tier
- Channel
- Venue

- Location
- Purchase Date
- Purchase Channel
- Original Price
- Promotion Used
- Usage Status

Ticket Status

Configure eligibility for:

- Valid
- Unused
- Partially Used
- Used
- Expired
- Cancelled
- Suspended

Example:

Upgrade allowed only before first admission.

Time-Based Eligibility

Examples:

Upgrade allowed until 2 hours before event start.

Annual-pass conversion allowed within 7 days of original visit.

Upgrade unavailable after ticket expiration.

Membership Conditions

Example:

Gold Member:

VIP Upgrade Eligible.

Standard Customer:

VIP Upgrade Not Available.

Transaction History

Rules may consider:

- Previous Upgrade
- Previous Conversion
- Refund History
- Redemption History

Rule Result

Return:

Eligible

or

Not Eligible

with reason.

Example:

Not eligible — ticket has already been redeemed.

Acceptance Condition

TICVAI can determine upgrade/conversion eligibility consistently using configurable ticket, customer, time and transaction conditions.

11.1.4 — Upgrade Timing, Usage & Ticket Status Rules

Purpose

Define how ticket lifecycle state affects upgrade and conversion behavior.

This deserves its own screen because a ticket may already have been partially consumed.

Usage Conditions

Support:

- Before First Use
- After First Use
- Partially Consumed
- Before Visit
- During Visit
- After Visit
- Before Expiry
- Grace Period

Example — Attraction Upgrade

Customer purchased:

1 Attraction Pass

and already visited Attraction A.

They request:

3 Attraction Pass

TICVAI should determine whether the consumed entitlement can be credited toward the upgraded product.

Remaining Entitlement

Configure treatment of:

- Used Admissions
- Unused Admissions

- Remaining Days
- Remaining Stored Value
- Remaining Benefits

Status Transformation

Example:

Original Ticket:

VALID

↓

Upgrade Completed

↓

Original Ticket:

SUPERSEDED

New Ticket:

VALID

Original Ticket Handling

Configure:

- Invalidate
- Supersede
- Retain for History
- Link to New Ticket
- Partially Retain Entitlement

Expired Tickets

Configure whether:

- Never Eligible
- Eligible Within Grace Period
- Supervisor Exception Allowed

Acceptance Condition

Upgrade and conversion rules correctly account for ticket lifecycle, usage and remaining entitlement before allowing a transaction.

11.1.5 — Upgrade Financial Treatment & Price Difference Rules

Purpose

Define **how the financial relationship between the old and new product should be treated.**

Important:

Area 11 defines the financial policy.

Area 10 performs the actual monetary calculation.

Financial Methods

Support:

Full Difference

Target Price – Eligible Original Value

Fixed Upgrade Fee

Example:

Standard → Premium = AED 100.

Percentage Upgrade

Example:

25% of target price.

Pro-Rata

Based on remaining validity/entitlement.

Credit-Based

Apply eligible original ticket value as credit.

No Credit

Customer pays full target price.

Complimentary Upgrade

Difference = zero where authorized.

Example

Original Ticket:

AED 250

Target Ticket:

AED 400

Eligible Credit:

AED 250

Upgrade Difference:

AED 150

Price Source

Configure whether target pricing uses:

- Current Selling Price
- Original-Date Price
- Upgrade-Specific Rate
- Contracted Rate
- Membership Rate
- Fixed Upgrade Price

Dynamic Pricing

This is important.

If the target ticket is dynamically priced:

Should the upgrade use the **current dynamic price**, a protected upgrade rate, or another configured rate?

This must be configurable.

Discounts

Configure whether existing:

- Promotion
- Membership Discount
- Voucher
- Corporate Discount

can be carried forward.

Area 10 Integration

Send calculation context to the central pricing engine.

Area 10 returns:

- Target Price
- Credit
- Price Difference
- Discount
- Fees
- Tax
- Rounding
- Final Amount

Acceptance Condition

Area 11 can define upgrade financial policies while all monetary calculations are resolved consistently through TICVAI's central pricing engine.

11.1.6 — Pro-Rata, Residual Value & Entitlement Credit Configuration

Purpose

Handle complex upgrades where part of the original product has already been consumed.

This is particularly important for:

- Multi-Day Passes
- Memberships
- Annual Passes
- Multi-Attraction Products
- Packages

- Stored Entitlements

Pro-Rata Methods

Support:

Time Based

Remaining Days / Original Days.

Usage Based

Remaining Uses / Total Uses.

Value Based

Remaining Commercial Value.

Entitlement Based

Value of unconsumed benefits.

Fixed Credit

Configured commercial amount.

Example

Original:

3-Day Pass = AED 600

Used:

1 Day

Remaining:

2 Days

Configured method:

Time-Based Pro-Rata

Eligible Credit:

AED 400

Target:

Annual Pass = AED 1,500

Amount before applicable fees/taxes:

AED 1,100

Credit Restrictions

Configure:

- Maximum Credit %
- Minimum Upgrade Amount
- Credit Expiry
- Non-Creditable Components
- Fee Exclusion
- Tax Treatment

Entitlement Mapping**Example:****Original Product:**

3 Attraction Admissions

Used:

1

Remaining:

2

Target:

Unlimited Annual Pass.

System determines how remaining entitlements are treated after conversion.

Negative Difference**If calculated target value is lower than eligible credit:****Configure:**

- No Refund
- Refund Difference
- Wallet Credit
- Voucher Credit
- Supervisor Approval

Actual refund execution is handled by the relevant payment/refund engine.

Acceptance Condition

TICVAI can correctly determine eligible residual value and entitlement credit for partially consumed products according to configurable commercial policies.

11.1.7 — Person-Type, Product & Entitlement Conversion Rules**Purpose**

Handle conversions that change more than simply the commercial level of a ticket.

Person-Type Conversions**Support:**

- Child → Adult
- Junior → Adult
- Senior → Adult
- Resident → Tourist
- Standard → Member
- Custom Person Types

Example

Child Ticket:

AED 180

Customer no longer qualifies for Child category.

Convert:

Child → Adult

Adult Current Price:

AED 250

Difference calculated through Area 10.

Product Conversion**Examples:****General Admission**

→ Fast Track

Single Attraction

→ Combo Pass

Day Pass

→ Annual Pass

Standard Membership

→ Premium Membership

Entitlement Conversion**The system must compare:****Existing Entitlements**

vs.

Target Entitlements**and identify:**

- Retained
- Replaced
- Added
- Removed
- Already Consumed

Validation**Example:**

Customer already consumed an entitlement that does not exist in the target product.

TICVAI should apply configured policy rather than silently converting.

Customer Qualification**A target product may require:**

- Age

- Membership
- Residency
- Corporate Association
- Identity Verification
- Other eligibility rules

Acceptance Condition

TICVAI supports governed conversion between person types, product types and entitlement structures while preserving eligibility and consumption integrity.

11.1.8 — Bulk, Group & Assisted Upgrade Operations

Purpose

Support operational upgrades involving multiple tickets rather than requiring staff to process each individually.

Use Cases

- Family Booking Upgrade
- Group Upgrade
- Corporate Booking
- School Group
- Tour Group
- Event Relocation
- Mass Service Recovery
- Venue Operational Change

Bulk Selection

Select by:

- Order
- Reservation
- Group
- Event
- Performance
- Ticket Type
- Seat Section
- Customer Segment

Bulk Actions

Support:

- Upgrade All
- Upgrade Selected

- Convert Product
- Change Person Type
- Apply Complimentary Upgrade
- Apply Fixed Upgrade
- Move to Alternative Performance

Eligibility Preview**Before execution:**

Selected: 150 Tickets

Eligible: 142

Not Eligible: 8

Reasons:

- 3 Already Used
- 2 Expired
- 3 Target Product Unavailable

Financial Preview**Show:**

Total Original Eligible Value

Total Target Value

Total Upgrade Difference

Fees

Taxes

Final Collection/Refund Requirement

Partial Processing**Configure:**

Process eligible tickets and exclude failures

or

Fail entire batch if any ticket is invalid.

Approval

High-value or complimentary bulk upgrades may invoke the central approval framework.

Acceptance Condition

Authorized operations teams can safely process large group and bulk upgrades with eligibility, financial and exception validation before execution.

11.1.9 — Upgrade Execution, Credential Regeneration & Channel Controls

Purpose

Control what happens operationally once an upgrade or conversion is approved and financially completed.

Execution Flow

Eligibility Passed

↓

Financial Calculation

↓

Payment / Refund if Required

↓

Upgrade Commit

↓

Original Ticket Updated

↓

New Entitlement Created

↓

Credential Updated

↓

Channels Synchronized

Credential Handling

Depending on configuration:

- Regenerate QR
- Invalidate Old QR
- Update Dynamic QR
- Update RFID
- Update NFC
- Update Wallet Pass
- Reissue Ticket
- Preserve Existing Credential

Security Principle

Where ticket rights materially change, the old credential should not remain independently usable.

Access Control Integration

Immediately synchronize:

- New Entitlement

- New Access Rights
- New Zone
- New Date
- New Performance
- Old Credential Invalidation

Channel Controls

Configure whether upgrade/conversion is available through:

- B2C Self-Service
- Mobile App
- POS
- Call Center
- Box Office
- Kiosk
- B2B
- Reseller
- API

Channel Differences

Example:

B2C:

Standard → Premium only.

Call Center:

Standard → Premium + Event Exchange.

Supervisor POS:

All authorized paths.

Confirmation

After execution generate:

- Updated Ticket
- Updated Receipt/Invoice
- Confirmation Email
- SMS/WhatsApp where configured
- Wallet Pass Update

Communication delivery should reuse TICVAI's central CRM/notification capabilities rather than creating a separate messaging engine here.

Failure Recovery

If payment succeeds but ticket conversion fails:

Transaction must enter controlled recovery/reconciliation rather than leaving payment and entitlement states inconsistent.

Acceptance Condition

Completed upgrades correctly update tickets, entitlements, credentials, access rights, financial records and applicable sales channels as one controlled transaction.

11.1.10 — Upgrade History, Exception Management & Audit Explorer

Purpose

Provide complete operational and financial traceability for every upgrade, downgrade, exchange and conversion.

Transaction History

Store:

- Transaction ID
- Original Ticket
- New Ticket
- Source Product
- Target Product
- Customer
- Order
- Transaction Type
- Original Value
- Eligible Credit
- Price Difference
- Fees
- Tax
- Payment/Refund
- Channel
- Agent
- Date/Time
- Rule
- Status

Conversion Chain

Visualize:

Standard Admission

→ Premium Admission

→ VIP Admission

→ Annual Pass

with each transaction linked.

Exception Queue

Display:

- Eligibility Override
- Financial Override
- Expired Ticket Exception
- Manual Credit
- Complimentary Upgrade
- Failed Credential Update
- Failed Payment Reconciliation
- Channel Synchronization Failure

Manual Override

Authorized users may override configured restrictions where policy permits.

Require:

- Reason
- User
- Role
- Approval
- Supporting Note
- Timestamp

Audit Trail

Capture:

Who → Changed What → From → To → Why → When → Approval → Result

Search

Search by:

- Ticket
- Order
- Customer
- Event
- Product
- Agent
- Channel
- Transaction Type
- Date
- Exception

AI Operational Analysis

AI can surface patterns such as:

38% of Premium upgrades occur within 48 hours of the visit.

or:

Upgrade failures increased on OTA-originated tickets due to missing target-product mapping.

AI is analytical here and should not bypass upgrade rules.

Acceptance Condition

Every upgrade/conversion transaction and exception can be fully reconstructed across commercial, financial, ticketing, credential and user activity.

Board 1 — Final Screen Register

#	Backend Screen	Core Responsibility
11.1.1	Upgrade & Conversion Command Center	Central operations
11.1.2	Upgrade & Conversion Path Builder	Source-to-target relationships
11.1.3	Upgrade Eligibility & Qualification Rules	Customer/ticket eligibility
11.1.4	Upgrade Timing, Usage & Ticket Status Rules	Lifecycle and usage controls
11.1.5	Upgrade Financial Treatment & Price Difference Rules	Financial policy
11.1.6	Pro-Rata, Residual Value & Entitlement Credit Configuration	Partial-value calculations
11.1.7	Person-Type, Product & Entitlement Conversion Rules	Complex conversions
11.1.8	Bulk, Group & Assisted Upgrade Operations	Mass operations
11.1.9	Upgrade Execution, Credential Regeneration & Channel Controls	Transaction execution
11.1.10	Upgrade History, Exception Management & Audit Explorer	Traceability

One adjustment to the Area 11 name

I recommend we keep the master area named:

Area 11 — Ticket Upgrade, Exchange & Conversion

even though the board is called **Upgrade & Conversion Management**.

That's because **exchange** is a genuine business operation and should remain explicit in the module architecture—for example:

Saturday 18:00 ticket → Sunday 18:00 ticket

or

Event A → Event B

But we don't need a separate Exchange screen because the same **path + eligibility + financial treatment + execution architecture** handles it cleanly.

So at this stage, I still believe **1 board / 10 screens is sufficient for Area 11**. The next step after freezing this SOW is to design **all 10 screens of Board 1**, following this register exactly.